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International market selection and subsidiary performance: A neural network approach

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ABSTRACT

How should multinational enterprises (MNEs) select international markets? We develop a model of international market selection that adds firm-specific advantages and transaction cost considerations to previously explored target market factors based on Dunning's Eclectic Framework. Results obtained using neural network (NN) analysis indicates that our model has strong predictive power in explaining international market selection. Further tests show that firms selecting international markets predicted by the model reported significantly higher subsidiary performance relative to firms whose investments were not predicted by the model. Our results provide strong initial evidence that a firm-level strategic approach to international market selection facilitates MNE success.

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1. Introduction

International market selection (the choice of foreign markets) is a critical component in the success or failure of multinational firms (Douglas & Craig, 1992). It is important because location can affect the appropriateness and applicability of firm-specific resource advantages (Priem & Butler, 2001), transaction costs (Brouthers, Brouthers, & Werner, 2003), and/or the ability of firms to effectively transfer knowledge (Kostova, 1999). We applied Dunning's *OLI framework*, to the issue of international market selection and asked the question: Can the Dunning framework predict superior subsidiary performance in international market selections?

Dunning suggests that firm location choice will depend upon the firm's ability to create and sustain some income-generating advantage or set of advantages over and above that of competitors or potential competitors (Dunning, 1993). He states: "firms will seek to site their value-added activities at the most profitable points in space" (Dunning, 2001, p. 177). We suggest that Dunning's OLI framework may help firms determine those most profitable "points in space" (countries) because it takes into account three primary drivers of firm success: location advantages, ownership advantages and internalization (efficiency) advantages.

The primary contribution of this study is that it extends the OLI framework (also known as the eclectic model) to a new venue. The study is the first to use Dunning's framework to select superior performing international markets. It attempts to advance the literature in two ways. First, we know very little about how international market selection (IMS) decisions are made and even less about how they "should" be made. Although "effective market choice is a strategic decision," (Papadopoulos, Chen, & Thomas, 2002, p. 166), little research has used existing

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theories/models to explain how firms choose international markets (Brouthers & Nakos, 2005). This study uses the OLI framework to predict international market selection. Second, no prior study has developed a normative model of international market selection. We posit that Dunning's OLI framework predicts which international markets will be selected. We also propose that firms that place subsidiary units in markets predicted by the framework, on average, perform better, in terms of sales growth, overall performance, firm reputation, and market access, than firms that do not. Thus, we hypothesize that with respect to international market selection the OLI framework is both predictive and normative. This is examined using a sample of large European Union (EU) MNEs selecting target host countries from among the emerging markets of Central and Eastern Europe.

2. Dunning's framework and subsidiary performance

Dunning's Eclectic Framework incorporates transaction costs, firm advantages and host country characteristics. The OLI framework (Dunning, 1988, 1993, 1995) was originally developed to explain the existence of various forms of foreign direct investment [FDI], including marketing seeking investment (Dunning, 1993). Implicit in the definition of FDI is the notion of location. His framework was, at least in part, intended to identify and evaluate what factors contribute to the choice of location (Dunning, 1988, 1993).

Dunning's framework intimates that firms choose international markets based on three sets of variables: *Ownership advantages*, consisting of firm-specific resources, experience and proprietary knowledge; *Location advantages*, which include market potential and market risk; and *Internalization advantages*, which are concerned with the reduction of transaction costs (Agarwal & Ramaswami, 1992; Dunning & McQueen, 1982). This framework has been used to examine numerous business strategy research questions, including economic development (Narula & Dunning, 2000), strategic alliances (Dunning, 1995), firm decision making, divestment by MNEs, the economics of technological innovation, entrepreneurship and change (Dunning, 1988, 1993), and as a predictive/normative model of international entry mode choice (Agarwal & Ramaswami, 1992; Brouthers, Brouthers, & Werner, 1999; Tse, Pan, & Au, 1997). Singh and Kundu (2002) have extended this framework to the domain of e-business. Clearly, the OLI framework has served as an effective catalyst for theory development and application (Cantwell & Narula, 2001). In similar fashion to the studies mentioned, this paper applies the OLI model to a new and separate research domain (market selection), testing the model's ability to make international market selections that result in superior firm performance.

Pantzalis (2001) suggests that the OLI framework is a means of explaining the "why", "where", and "how" of the overseas activity of firms (with the "where" describing international market selection). However, previous work using the framework mentions international market selection only in passing (i.e. Sethi, Guisinger, Phelan, & Berg, 2003). We develop an explanation of how using OLI variables to make international market selections can result in superior subsidiary performance.

2.1. Location advantages

A firm's selection of an international market is based on a number of location-specific factors. Narula and Dunning (2000, p. 160) note that MNEs "are being forced to give more attention to the availability and quality of the largely *L*-specific created assets of alternative investment sites. Indeed, choosing the 'right' portfolio of locations for their value-added activities is a competitive advantage in its own right". In another article Dunning (1998) suggests that MNEs seek locations where they can use their core competencies most efficiently. Hence, Dunning appears to suggest that the OLI framework and host country factors, such as market potential, growth potential, political, social and economic stability, and political risk, play a major role in international market selection (Narula & Dunning, 2000).

Previous international market selection research has noted that location-specific advantages are incorporated into IMS decisions because they take into consideration the market potential of a given location as well as the governmental and economic stability of the market (Douglas & Craig, 1982; Papadopoulos & Denis, 1988). According to Cavusgil and Zou (1994) selecting countries with greater market potential and stability allows firms to achieve greater subsidiary performance. Further, as Makino, Lau and Yeh (2002) suggest, when location advantages can be combined with firm-specific advantages, firms may be able to generate new or stronger competencies that provide an advantage over competitors; these advantages may lead to superior performance outcomes for subsidiaries. Thus, previous research suggests that the inclusion of location-specific advantages in the IMS decision helps firms select markets where superior performance may be achieved (Cavusgil & Zou, 1994; Makino et al., 2002; Papadopoulos et al., 2002).

2.2. Ownership advantages

Dunning (1988, p. 2) states, ownership advantages "... sometimes called competitive or monopolistic advantages—must be sufficient to compensate for the costs of setting up and operating a foreign value-adding operation, in addition to those faced by indigenous producers or potential producers." Ownership advantages, which include elements such as intangible assets, management skills and trademarks (Dunning, 1988, 1993), may influence IMS in two ways. First, firms possessing mobile ownership advantages prefer to enter markets in which these specific advantages can be effectively exploited (Makino et al., 2002). Second, firms seeking additional ownership advantages may select markets where location-bound advantages can be combined with firm-specific advantages to increase the overall value of O-advantages (Dunning, 1998; Makino et al., 2002). Thus, it appears that MNE ownership advantages may also influence the international market selection decision.

Considering ownership or firm-specific advantages in the international market selection decision may also lead

to the selection of markets where firms perform well. Firm-specific advantages tend to be context specific and may be more appropriate in one market than in another (Madhok, 1997; Priem & Butler, 2001). Aligning firm-specific advantages with context may provide a firm with competitive advantages that result in superior firm performance (Madhok, 1997; Priem & Butler, 2001).

2.3. Ownership–location interactions

While it is tempting to treat ownership and location advantages as distinct elements in the selection of international markets, Dunning intimates that they may actually be interdependent. Location-based advantages interact with ownership advantages. He states, "... a firm's propensity to invest in a particular country is often strongly influenced by the factor endowments, created capabilities and markets available in that country relative to others, as well as the extent to which it is perceived that the economic system and policies of a country enable it to exploit its O advantages profitably" (Dunning, 1993, p. 270). What Dunning is suggesting is that selecting a particular market may enhance or undermine the specific ownership (O) advantages of a firm. Thus, in addition to the main effect influences of ownership and location advantages, there may be an interaction between the two that influences international market selection.

2.4. Internalization advantages

Internalization advantages (which can also be thought of as efficiency advantages) tend to vary from market to market with the focus on the boundaries of the firm and the decision to internalize market based transactions (Dunning, 1993; Hill, 1990; Williamson, 1985). For example, in Japan little incentive exists for firms to internalize markets in order to avoid the costs of broken contracts or to ensure the quality of products/services (Dunning, 1995). One reason that a firm selects a host country market is that it allows the transfer of ownership advantages so efficiently that the internalization advantages overcome the competitive disadvantage the firm experiences *vis-á-vis* local firms—the so called "liability of foreignness" (Zaheer, 1995). Internalization advantages also focus on minimizing transaction costs in different locations. Transaction costs influence the efficiency of a firm's operations (Williamson, 1985). By selecting a market in which transaction costs are low, firm performance is enhanced (Brouthers et al., 2003).

In summary, our theoretical discussion suggests that a model of international market selection that incorporates target market (location-specific) factors with firm-specific and internalization advantage factors will do a superior job of predicting international market selection. The reason for this is that countries differ from each other with respect to location advantages and internalization costs while firms differ from each other with respect to ownership advantages and internalization costs.

Country-specific opportunities for individual firms depend at least in part on firm-specific ownership advantages meshing with variations in national location advantages; firm-specific ownership advantages may have greater/lesser value depending on specific host country environments. Further, both firm and country differences influence the level of internalization advantages for each firm/country combination. Based on our review, we conclude that international market selection may be a process of balancing differences between countries and firms so that they provide the most effective combination of ownership, location, and internationalization advantages.

2.5. Normative market selection

It also appears that Dunning's framework may offer a normative approach to selecting superior performing international markets. In contrast, previous tests of international market selection tend to be descriptive in nature, examining how well a firm's IMS could be predicted using a set of independent variables (usually target country level variables) without exploring performance implications (see Papadopoulos et al., 2002). Yet research, such as Shaver (1998) and Brouthers et al. (2003) posits that in order for international business theory to be helpful to managers it needs to move beyond mere description and attempt to help managers improve international firm performance.

Can the Dunning framework predict superior performing international market selections? Based on our logic we conclude that using all of the OLI dimensions in IMS results in the selection of superior performing international markets. Location-specific advantages take into consideration market potential as well as the governmental and economic stability of the market (Douglas & Craig, 1982; Papadopoulos & Denis, 1988), both of which have been found to contribute to superior firm performance (Cavusgil & Zou, 1994). Ownership advantages placed in applicable and appropriate markets provide firms with competitive advantages that result in superior firm performance (Madhok, 1997; Priem & Butler, 2001). Finally, internalization advantages improve the efficiency of a firm's operations. Markets where transaction costs are low enhance firm performance (Brouthers et al., 2003). Thus, Dunning's OLI framework may provide a normative model of international market selection, identifying location, ownership and internalization factors that result in market selection decisions that lead to superior firm performance. Based on our discussion, we hypothesize:

Hypothesis 1. MNEs whose international market selections were predicted based on their OLI advantages will have higher performance (in terms of sales growth, overall performance, firm reputation, and market access) at the subsidiary level than those that select international markets not predicted by their OLI advantages.

3. Methodology

We used a two-step methodology to examine the predictive and normative utility of Dunning's OLI framework for international market selection. First, we used a Multi-Layered Perceptron (MLP) NN trained by a back-

propagation (BP) algorithm to assess the effectiveness of the OLI variables in predicting which international markets our sample of firms selected. Second, we used hierarchical regression to determine whether the firms that selected markets predicted by our OLI model had significantly higher levels of subsidiary performance than firms that selected misaligned (non-predicted) markets. MLP analysis is used to determine the optimal pattern of input/output relationships (Thieme, Song, & Calantone, 2000). MLP is widely used in discriminating among nominal level variables (Thieme et al., 2000). MLP offers a few advantages over linear forms of analysis like discriminant or multinomial logistic regression analysis. MLP is flexible and can easily deal with nonlinear data (Bishop, 1995; Ripley, 1996). MLP requires no *a priori* knowledge of relationships or distributional assumptions about the data (Veiga, Lubatkin, Calori, Very, & Tung, 2000). Additionally, scholars (Sharda, 1994; Thieme et al., 2000) have empirically demonstrated that NN commonly outperforms classical linear statistical techniques like regression or discriminant analysis. MLP is a good choice when the inputs are relevant and fairly large and most of the predictive information lies in a low-dimensional subspace of the original dimension of the input data (Michie, Spiegelhalter, & Taylor, 1994).

In such a hyper-separation surface MLP can obtain a good fit in the training sample by memorizing each sample point location. However, the calibrated model may not subsequently work on new samples (generalization). To ensure that the MLP model generalizes we divided our analysis into two steps. In the first step we randomly chose sample points to calibrate the OLI paradigm. In step 2, we linked the calibrated paradigm to performance with statistical significance. Step 2 data (performance attributes) did not participate in creating the OLI paradigm in step 1. Therefore, the method attempts to prove the hypothesis through generalization.

3.1. Sample

A sample of Dutch, British and German firms doing business in Central and Eastern Europe (CEE) were used in this study. CEE was used as the research site because countries in this region had similar experiences with centrally planned economies during the post-war period followed by widely differing political and market transitions in the 1990s. Before 1989 cross-border economic activity in the CEE was primarily restricted to other Eastern Bloc countries. Since that time these countries have made considerable progress in their integration into the global economy, particularly in relation to the European Union (Cernat & Vranceanu, 2002). This allowed us to examine market-seeking IMS decisions in the context of countries in the initial stages of opening their economies to multinational enterprises, thereby minimizing legacy effects. Hence, firms from countries in the EU with the largest investments in CEE (Netherlands, the U.K., and Germany) were used in this study. Our paper is restricted to first-time entries into a given host country market.

We compiled our own listing of EU firms with CEE business activity because no list of such firms was available. Our sample came from four sources: (1) 122 Dutch firms from the REACH CD-ROM database of firms doing business in CEE; (2) 297 Dutch firms that had participated in a seminar about doing business in CEE; (3) 400 German and 371 British large companies from the AMADEUS CD-ROM (we assumed that larger companies were more likely to engage in commerce in the CEE). Thus, a total of 1190 Dutch, German and British firms were surveyed[2].

Managers were asked about each company's most recent international market selection decision in Central and Eastern Europe. They were specifically asked to identify *market-seeking* IMS (as opposed to *labor-seeking* or *resource-seeking* IMS). From the 1190 surveys sent to firm headquarters, we received 339 responses for an initial response rate of 28.4%. Of the 339 responses 274 were sufficiently complete to be used in the first step of the analysis (see Table 2 for responses by country). In the second step of the analysis, the number of responses was reduced to 197 because the remainder of the firms did not complete the performance measures section of the survey. The second step sample included companies with foreign market operations in the Czech Republic ($n = 38$), Russia ($n = 69$), Poland ($n = 25$), Hungary ($n = 52$), and Ukraine ($n = 13$).

3.2. Dependent variables

In the first step of our analysis five categories of nominal dependent variables were employed to examine the predictive ability of the OLI framework in international market selection. These categories were (1) Czech Republic; (2) Poland; (3) Hungary; (4) Russia and (5) Ukraine.

In the second step, two dependent performance variables were used to examine the normative ability of the OLI framework. As in prior efforts (Brouthers, 2002; Makino & Beamish, 1998), this study used perceptual measures of firm performance rather than objective measures. Previous research has found that perceptual and objective measures of firm performance tend to be highly correlated (Dess & Robinson, 1984; Geringer & Hebert, 1991; Kim & Slocum, 2008). Past research has also found two major problems in attempting to use "objective" financial measures in international business research. First, managers are willing to provide perceptual performance measures but are reluctant to provide financial measures of performance, especially at the subsidiary level. Second, cross-national financial comparisons are problematic due to differences in financial and accounting standards and/or the effects of exchange rate fluctuations (Brouthers & Xu, 2002). Thus, perceptual measures were used to evaluate subsidiary performance.

Two dependent variables were constructed. The first, *quantitative* performance, was treated as a multidimensional construct using two "quantitative" performance

[2] Data was collected through a survey mailed to the sample firms. As determined in discussions with potential survey participants, survey questionnaires were directed to the Vice President of Central and Eastern European operations in each firm. Hence, as in previous research (Agarwal and Ramaswami, 1992; Brouthers, 2002) we utilize key informants to gather our data. Three mailings were made for each group of firms and questionnaires were translated into German for the German companies.

dimensions: sales growth and overall performance. The second dependent variable, *qualitative* performance, was also a multidimensional construct consisting of two "qualitative" performance dimensions: firm reputation and market access. For each performance measure respondents were asked, "How satisfied are you with this subsidiary's performance as measured by ..." (the performance dimensions). Responses were on a 10-point scale ranging from 1 (very dissatisfied) to 10 (very satisfied). Factor analysis confirmed that two factors (that we termed quantitative and qualitative performance) were created by these four performance measures (Cronbach's Alpha equaled $\alpha = 0.72$ and $\alpha = 0.76$, respectively).

3.3. Independent and control variables

Independent variables consisted of multiple dimensions based on previous operationalizations of the OLI advantages found in the seminal work of Agarwal and Ramaswami (1992) and employed in subsequent studies (Brouthers et al., 1999; Driscoll & Paliwoda, 1997) as discussed below (detailed measures are provided in the Appendix A).

3.3.1. Ownership advantages

Four variables were used to measure ownership advantages: *worldwide sales, experience, R&D intensity*, and *creative rating* (Agarwal & Ramaswami, 1992; Brouthers et al., 1999). Using a continuous ratio measure, respondents were asked to provide sales volume information based on their *worldwide sales* (the firm's sales volume). *Experience* was measured using three items: sales intensity (sales outside the domestic market divided by firm sales volume), experience outside the home country in number of years, and experience outside the home country by number of countries. Since sales intensity was measured differently than the other two, the three items were combined by running a factor analysis that used principal component loading (Gnanadesikan, 1977; Seal, 1964). Factor analysis confirmed that these three items loaded into a single factor (Cronbach's alpha equaled 0.70). *R&D intensity* was measured as R&D spending divided by firm sales volume. In addition, a 7-point bi-polar scale was used to determine a *creativeness rating* for each firm (Agarwal & Ramaswami, 1992). This rating was intended to assess the ability of the firms to develop new and creative products and services.

3.3.2. Location-specific advantages

Location-specific advantages were measured using four questions to assess the attractiveness of international markets (Agarwal & Ramaswami, 1992; Brouthers et al., 1999). *Market Potential* was based on perceived managerial assessment of market size. *Growth Potential* consisted of perceived managerial assessment of growth potential. The variable *General Stability* measured managerial perceptions of the stability of the political, social, and economic conditions of the host country. Managerial perceptions of the host government's policies toward conversion and repatriation of profits was measured with the variable *Government Risk*. A 7-point bi-polar scale was used for each question. A factor analysis showed that the first two items (market and growth potential) loaded on a single factor, *location1* (Cronbach's alpha equaled $\alpha = 0.70$). This factor was multiplied by the four ownership variables to create four ownership/location interaction variables. Each of the other two location-specific variables (government risk and general stability) loaded on separate factors and for that reason each was multiplied by the four ownership variables to create eight more ownership/location interaction variables.

3.3.3. Internalization advantages

Two items were used to assess internalization advantages (Agarwal & Ramaswami, 1992; Brouthers et al., 1999). The *Costs of Contracts* variable asked about how managers perceived the costs of making and enforcing contracts in the selected international market compared to their home country. The second item, *Risk of Dissemination*, consisted of perceived managerial risk of the dissipation of proprietary knowledge. Both of these items used a 7-point bi-polar scale.

3.3.4. Fit

A variable labeled "Fit" served as the independent variable of interest for assessing the performance implications of using the OLI framework for international market selection. The Fit variable was determined by the MLP analysis (see our analysis section). Fit was coded 1 if the MLP analysis determined that the OLI variables correctly predicted the firm's international market selection. Alternatively, Fit was coded 0 if the MLP analysis established that the OLI variables failed to predict the international market choice.

3.3.5. Industry type

One control variable was included in the analysis, *industry type* (Brouthers, 2002). Respondent firms were asked to indicate whether they were making a manufacturing or service based market entry. As in previous research, a dichotomous industry type variable was created where manufacturing firms were assigned a value of zero and service firms were assigned a value of one.

In instances in which independent and dependent variables originate from a single respondent, there is a concern that common method variance can be a problem (Hair, Anderson, Tatham, & Black, 1992). The following features, which have been found to lessen common methods variance, were built into our questionnaire: (1) questions operationalizing the OLI model were objective, not perceptual in nature; (2) the anonymity of respondents was protected (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003); (3) each item was worded carefully (Torangeau, Rips, & Rasinski, 2000); (4) criterion items and predictor items were separated in the instruments (Parkhe, 1993); and (5) common methods variance was not found to be a problem when we used the one factor test recommended by Podsakoff and Organ (1986); we had 14 factors, the highest of which explained only 24.22% of the variance.

3.4. MLP and regression analysis

We employed a categorical analysis using an MLP network trained by a back-propagation (BP) algorithm

(Rumelhart, Hinton, & Williams, 1988) to examine the predictive ability of the OLI framework to determine international market selection. Initially we tried a few training techniques, such as Levenberg-Marquardt, quasi-Newton, conjugate gradient and back-propagation (BP). We obtained very similar results indicating that the techniques chosen had no impact on the results. We, therefore, chose BP, which is well-referenced in the literature.

The MLP network has a layer of input nodes (independent variables), one or more layers of hidden nodes, and a layer of output nodes (dependent variables). The nodes of the hidden layers connect the input layer nodes with the output layer nodes (Thieme et al., 2000). We designed the architecture based on the principle that a smaller net is better than a large one (Xiang, Ding, & Lee, 2005). The number of hidden nodes should equal at least the number of input variables in order to capture 70–90% of the variance on the input data (Boger & Guterman, 1997). We therefore used one hidden layer, starting with eight hidden nodes, increasing the number to 15 to achieve the best performance. We used linear combination function and logistic activation function to input to the hidden nodes to an output value from the hidden nodes.

The BP algorithm training method extracts the functional relationship between values of the input variables and the output variables, through assignments of weights (Veiga et al., 2000). To perform the MLP analysis, data were split into two sets, one for training the neural net and one for testing the calibrated model (Veiga et al., 2000). The splitting parameter (training sample percent) was chosen to be 80 through random sampling among the set of 50, 60, 70 and 80 because of its superior performance. During training, the outputs of a neural net come to approximate target values given the inputs in the training set. An MLP uses the information available from the independent variables for each observation in the training set to compute an output value. The output value is then compared with the target value to generate error signals for all units. There will be no error signal if there is no difference between the two values. Otherwise, the training involves a backward pass through the network, during which error signals are sent to all units in the network. Weight changes in network connections are proportional to the error signals. In this procedure the constant of proportionality is called the "learning rate." The larger this constant the larger the changes in the weights in each step. The network continues the BP process until the sum-squared error is minimized over the sample patterns. Following Rumelhart et al. (1988) a learning rate of 0.1 and a momentum factor of 0.9 were used[3].

[3] The learning process can be rapid if the learning rate can be increased. However, the learning process can jump back and forth in the solution space if the learning rate is too high. This phenomenon is called "oscillation." One way to increase the learning rate without leading to oscillation is to include a "momentum" term in the weight change formula. It is a constant that determines the effect of past weight changes on the current direction of movement in weight space. This provides a kind of momentum in the weight space that effectively filters out high-frequency variations of the error-surface in the weight space.

The model was then qualified on the test set. The purpose was to determine how well the network made predictions for cases not in the training set. There are often instances where a NN model that works very well on the training set does not perform well on the test set. One reason may be that the calibration process memorized input and output locations in the problem space through repetitive presentations of samples. This faulty repetitive process of NN training is called "over-fitting" or "memorization" (Wieland & Leighton, 1988). In other words, "memorization" in NN corresponds to learning many specific input-to-output relations without generalizing between similar input/output pairs. If memorization occurs, the location-specific memory fails when the locations (relationships) of test samples change considerably with respect to the training sample. Generalization on the other hand, produces the proper relationship between the input and output variables.

To avoid "over-fitting" or "memorization", several techniques have been developed. We used the "early stopping" method suggested by Sarle (1995) to estimate the generalization error from the available training sample. The process took 350 iterations to achieve the minimum error in the test set. We used the values of the network weights after completion of 350 iterations to determine the hit rates of the model (for the five target countries) and in computing the values (0 or 1) of the variable we labeled as "*Fit*." We also ran an MLP analysis on the same data using only the country level (location) variables for comparative purposes.

In the second step of our analysis, hierarchical regression was used to determine whether "Fit" firms (firms whose market selections were predicted by the OLI model) had higher mean performance than "non-Fit" firms (firms whose market selections were not predicted by the OLI model). Shaver (1998) suggests that studies linking strategic decisions (in our case international market selection) with firm performance should compare and contrast performance outcomes between decisions predicted by theory (in our case the OLI variables) and those that result from determinations made outside the boundaries of established theory. This means it is important to examine both international market selection criteria as well as the linkage between that criteria and subsequent firm performance.

For each of our two performance dependent variables (qualitative and quantitative performance), two hierarchical regression models were tested. The first model uses OLI variables to predict firm performance. The second model adds our dichotomous "*Fit*" variable (coded 1 if the MLP model correctly predicted IMS, and coded 0 if it did not) to the first model in an effort to determine whether firms whose IMS can be predicted by the MLP using OLI variables (and their interactions) perform better than firms for whom IMS cannot be predicted, controlling for factors previous research has found to be linked to performance.

Fit was added hierarchically into a regression model in order to determine whether it contributed to explaining subsidiary performance above linear OLI

Table 1
Correlation matrix

	1	2	3	4	5	6	7	8	9	10	11	12	13
Mean (S.D.)	6.55 (2.2)	7.35 (2.0)	7.58 (15.2)	0.00 (1.0)	4.41 (5.2)	5.31 (1.2)	4.28 (1.5)	5.73 (1.4)	0.00 (1.0)	4.47 (1.8)	4.14 (1.7)	0.32 (.47)	0.64 (.48)
1. Quantitative performance	–												
2. Qualitative performance	.64*	–											
3. WW sales	.08	.04	–										
4. Experience	.03	.09	.27*	–									
5. R&D intensity	.10	.13	.32*	.26*	–								
6. Creative rating	.11	.08	.00	.11	.22*	–							
7. General stability	.21*	.26*	−.05	−.16	−.01	−.10	–						
8. Government risk	.13	.17	−.01	−.08	−.01	.01	.36*	–					
9. Location1	.14	.03	−.06	.05	.11	.18	.08	.06	–				
10. Cost of contracts	.23*	.18	.02	−.23*	−.10	−.24*	.38*	.24*	−.02	–			
11. Dissem. risk	.11	.32*	−.08	−.08	.16	.11	.41*	.23*	.15	.38*	–		
12. Industry type	.00	−.04	−.17	−.33*	−.19	.02	−.15	.03	−.07	−.07	−.09	–	
13. FIT	.21*	.19	.10	.06	−.03	.02	−.09	−.04	.12	.00	−.06	.14	–

* $p < 0.01$.

variable predictions. Because a *nonlinear* use of the OLI variables helped to create the *Fit* variable, we were concerned about colinearity in the hierarchical *linear* regression equation. Table 1 provides descriptive statistics as well as inter-correlations between the variables used in our study. No large correlations were observed between any of the independent variables. Additionally, the maximum VIF for the independent variables in our regression models was 1.48 (for the cost of contracts variable), substantially smaller than the maximum limit of 10 specified by Neter, Kutner, Nachtsheim, & Wasserman (1996). These results confirmed that multicollinearity was not a concern in the hierarchical regressions.

4. Results

The number of target country entries and hit rates for the complete OLI model are summarized in Table 2.

Table 2 indicates that the OLI model correctly classified 64% of the cases, significantly greater than the 20% classification rate that would have occurred without any prior information (chance rate). The OLI model correctly classified 60.3% of firms for the Czech Republic, 58.8% for Russia, 86.8% for Poland, 56.0% for Hungary, and 83.3% for the Ukraine. This represents a prediction rate that is more than 80% higher than the rate based only on target country specific (location) variables and over three times the chance rate.

The following hypothesis test (Marcoulides & Hershberger, 1997) indicated that the OLI framework hit rate significantly surpassed the hit rate obtained using only target market based measures: H_0 was the null hypothesis which stated that the OLI model hit rate did not surpass significantly the hit rate expected by a model using only location based variables, test statistic $(z) = (HF_0 - HF_e)/(HF_0(N - HF_e)/N)^{0.5}$, HF_0 = hit rate frequency observed in the sample by OLI model, HF_e = hit rate frequency expected by only L variables, n = number of case in our sample. The $z = 7.41$ for the sample rejected the null at $p < 0.01$. These findings strongly suggest that the OLI framework does a better job of predicting firms' international market selections than does an approach using only target market based measures.

4.1. IMS and performance

Next, we compared the mean performance for the "fit" (correctly predicted target markets) and "non-fit" firms. Because direction was hypothesized, a one tailed test was used to examine whether firms making international market selection decisions predicted by our model reported greater performance than other firms. The test statistic is given by:

$$\text{T.S.} = \frac{(\text{Difference in the mean performance ratings of the fit and non-fit firms})}{\text{NUM}}$$

where NUM = $\sqrt{(\text{var1}/n1) + (\text{var2}/n2)}$, var1, var2 are the sample variances, and $n1$ and $n2$ are number of samples of the fit and non-fit firms, respectively.

We found that firms whose IMS choices were predicted by the model (we labeled these "fit" firms) had a

Table 2
Predicting international market selection

Country of entry	Cases	Prior probability (percentage)	Hit ratio (OLI model percentage)
Czech Republic	53	19.3	60.3
Russia	90	32.8	58.8
Poland	38	13.8	86.8
Hungary	75	27.3	56.0
Ukraine	18	6.5	83.3
Total/average	274	20.0	64.0

Note: 64% of original grouped cases correctly classified by OLI variables.

quantitative performance mean of 6.64 compared to 6.36 for firms selecting markets not predicted (non-fit firms) by the OLI MLP analysis. For *qualitative* performance the mean was 7.45 for firms whose IMS choices were predicted by the model (fit firms) compared to 7.12 for non-fit firms. The improvements in quantitative and qualitative performances due to OLI advantage are 4.21% and 4.63%, respectively. While a 4–5% increase in explained performance may appear to be modest, such an improvement could translate into substantial revenues, depending on the size of the firm. Moreover, both differences were statistically significant ($p < 0.05$), providing empirical support for our hypothesis: *MNEs whose international market selections were predicted based on their OLI advantages will have higher performance (in terms of sales growth, overall performance, firm reputation, and market access) at the subsidiary level than those that select international markets not predicted by their OLI advantages.*

4.2. Hierarchical regression

Table 3 reports the results of hierarchical regression models for the dependent variable *quantitative performance*. Model 1 included all of the OLI variables and the control variable, *industry type*. This model significantly predicted *quantitative performance* ($p < 0.05$), and explained 6% of the variation in the dependent variable (adjusted R^2 of 0.06).

Model 2 added our dichotomous "Fit" variable to Model 1. This model was also significant ($p < 0.01$), with an adjusted R^2 of 0.10. The inclusion of the Fit variable (firms whose IMS were predicted by our OLI variables) significantly increased the adjusted R^2 ($p < 0.01$) from 6% to 9%.

In addition the Fit variable was significantly related ($p < 0.01$) to *quantitative performance* with "fit" firms reporting higher performance than "non-fit" firms, providing support for our hypothesis.

Model 3 added two home country dummy variables – Dutch (value is 1 if the home country is "Dutch" otherwise the value is 0) and German (value is 1 if the home country is "German" otherwise the value is 0). Results indicate that the fit variable remains statistically significant after adding the two dummy variables; neither dummy variable is statistically significant.

Table 4 reports the results of hierarchical regression models for our second performance dependent variable, *qualitative performance*. Model 1 included all of the OLI variables along with the control variable *industry type*. The OLI variables explain 9% of performance. Model 2 added our dichotomous "Fit" variable to Model 1.When the fit variable is added explanatory power increases from 9% to 13%, an increase of 44%.

In addition, the fit variable was significantly related ($p < 0.01$) to *qualitative performance* with "fit" firms reporting higher performance than "non-fit" firms. Model 3 again added two home country dummy variables to estimate qualitative performance – Dutch (value is 1 if the home country is "Dutch" otherwise the value is 0) and German (value is 1 if the home country is "German" otherwise the value is 0). Results indicate that the fit variable remains statistically significant after adding the

Table 3
Regression estimates for *quantitative* performance

	Model 1	Model 2	Model 3
FIT[a]		2.01***	1.91**
Ownership advantages			
Worldwide sales	0.00	0.00	0.00
Experience	0.21	0.16	0.17
R&D intensity	0.04	0.05	0.04*
Creative rating	0.28*	0.28*	0.28
Location advantages			
Location1	0.25	0.18	0.03
General stability	0.29*	0.31*	0.29
Government risk	0.03	0.04	0.04
Internalization advantages			
Cost of contracts	0.39**	0.37***	0.38
Dissemination risk	−0.13	−0.11	−0.10
Country dummy variables			
The Netherlands			0.85
Germany			0.72
Industry type	0.48	0.30	0.30
Constant	2.10	0.18	−0.60
R^2	0.14	0.17	0.18
Adjusted R^2	0.06	0.10	0.09
Model F	1.89**	2.24***	1.94**
Change in R^2		0.04***	0.01**
Change in adjusted R^2		0.03***	−0.01**

$N = 197$.
[a] The Fit variable was determined by the MLP analysis. Fit was coded 1 if the MLP analysis determined that the OLI variables correctly predicted the firm's international market selection and coded 0 if the MLP analysis established that the OLI variables failed to predict the international market choice.
* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.

Table 4
Regression estimates for *qualitative* performance

	Model 1	Model 2	Model 3
FIT		1.85***	1.82**
Ownership advantages			
Worldwide sales	0.00	0.00	0.00
Experience	0.30	0.26	0.26
R&D intensity	0.03	0.03	0.03
Creative rating	0.10	0.09	0.09
Location advantages			
Location1	−0.08	−0.15	−0.13
General stability	0.20	0.22	0.20
Government risk	0.07	0.08	0.08**
Internalization advantages			
Cost of contracts	0.11	0.08	0.08
Dissemination risk	0.26**	0.27***	0.27
Country dummy variables			
The Netherlands			0.33
Germany			0.10
Industry type	0.27	0.10	0.10
Constant	3.76***	1.99	1.78
R^2	0.16	0.20	0.21
Adjusted R^2	0.09	0.13	0.12
Model F	2.25***	2.73***	2.33***
Change in R^2		0.04***	0.01***
Change in adjusted R^2		0.04***	−0.01***

$N = 197$.
** $p < 0.05$.
*** $p < 0.01$.

two dummy variables; neither dummy variable is statistically significant.

The changes in explanatory power in both Table 3 and Table 4 were significant at the $p < 0.01$ level, which, given our sample size, is comparable to previously published studies on strategic fit (e.g. Griffith & Myers, 2005; Xu, Cavusgil, & White, 2006). Both sets of findings (Tables 3 and 4) provided strong empirical support for our hypothesis.

We also ran a regression analysis by industry type. Table 5 reports the regression results for both dependent variables—quantitative (Model 1) and qualitative performances (Model 2). The fit variable is statistically significant for both models indicating that our hypothesis is supported across industry types.

Table 5
Regression estimates for *quantitative* (Model 1) and *qualitative* performance (Model 2) by industry type

	Manufacturing		Services	
	Model 1	Model 2	Model 1	Model 2
FIT	1.95**	1.81**	1.99**	1.77**
Ownership advantages				
Worldwide sales	0.00	0.00	0.00	0.00
Experience	0.25	0.30	0.28	0.40
R&D intensity	0.06	0.02	0.04	0.00
Creative rating	0.36*	−0.09	0.36	−0.15
Location advantages				
Location1	0.03	−0.08	0.03	−0.12
General stability	0.25	0.13	0.29	0.11
Government risk	0.12	0.07	−0.19	0.12
Internalization advantages				
Cost of contracts	0.62***	0.17	0.57***	0.13
Dissemination risk	−0.04	0.30**	−0.07	0.37**
Constant	−0.34	2.80*	0.03	3.62**
R^2	0.24	0.21	0.21	0.19
Adjusted R^2	0.16	0.12	0.12	0.09
Model F	2.87***	2.46***	2.30**	2.01**

$N = 104$ and 93 for manufacturing and services, respectively.
* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.

5. Discussion and conclusions

Despite the centrality of international market selection to international marketing strategy, previous scholarship (Papadopoulos et al., 2002) suggests that little research has used existing theories/models to predict how firms choose international markets or evaluate the normative merits of IMS. Previous international market selection literature tends to be descriptive and focus on target country characteristics (e.g., Cavusgil, 1997; Ghemawat, 2001; Kumar, Stam, & Joachimsthaler, 1994; Papadopoulos & Denis, 1988; Wood & Robertson, 2000). It has largely ignored strategic factors such as firm-specific (ownership) and/or efficiency (internalization) advantages. Thus, host country factors, the "L" of the OLI framework, have been the primary focus of previous IMS scholarship. In contrast, this paper examined firm location choice in the context of strategic considerations associated with ownership and internalization advantages, as well as considering target country characteristics. Furthermore, this study extends previous research by examining the performance consequences of international market selection providing a predictive solution to managerial choice.

International market selection is not international entry mode choice. Mode choice deals with the level and type of investment (wholly-owned, joint venture, exporting). International market selection deals with picking a host country target market. Although there has been much theoretical and empirical examination of entry mode choice (Agarwal & Ramaswami, 1992; Brouthers, 2002; Brouthers & Hennart, 2007; Brown, Dev, & Zhou, 2003; Erramilli, Agarwal, & Dev, 2002; Kumar & Subramaniam, 1997; Zhao, Luo, & Suh, 2004), this has not been the case for international market selection. Neither is international market selection what the Uppsala school calls "internationalization" (the process by which firms evolve from being domestic to becoming international). It is not about the process of becoming an MNE; it is about how MNEs select host country target markets.

We attempted to advance international market selection scholarship in two ways. First, we employed Dunning's OLI framework to predict EU firms' choices of CEE markets; extending an existing model to a new domain. Second, we examined the normative merits of using Dunning's OLI framework to predict international market selection decisions. In order to be useful to

managers our model needed to provide not merely superior predictive ability, but be normatively superior as well. Hence, we attempted to advance the international marketing literature by using the eclectic framework as both a predictive and normative model of international market selection.

We suggest that Dunning's framework is especially appropriate for international market selection because it includes an assessment of internalization and ownership advantages (in addition to the traditional target market factors) that afford firms the opportunity to determine which markets they can most efficiently transfer their ownership advantages, thereby minimizing opportunity costs. Failure to evaluate whether or not a firm has an internalization advantage in a given market can undermine the ownership advantages of the firm and lead to problems commonly associated with liability of foreignness issues (Zaheer, 1995).

We found strong support for using Dunning's framework as both a predictive and normative model of international market selection, at least for market-seeking investment. We found that, on average, 65% of our sample's international market selections were classified correctly using the OLI variables. This is significantly higher than the 35% classified correctly using only target country level variables. Moreover, our results showed that firms that selected international markets based on their OLI advantages, on average, had higher levels of performance than firms that did not. Thus, based on our findings, we conclude that Dunning's framework appears to be superior to previous IMS methods, both as a predictive and normative model, at least for large European Union firms investing in Central and Eastern Europe.

5.1. Limitations and future research

This study has several limitations. First, no lists of Dutch, British, or German firms doing business in CEE could be located; as a result we attempted to develop a logical sampling frame. The responses showed that large numbers of firms surveyed did not enter CEE, making it impossible to compare respondents and non-respondents (we could not determine which firms that had not responded had entered CEE markets). Thus, we do not know if our sample is truly representative. Second, we examined only Dutch, British and German firms. For this reason our results may not generalize to firms from other nations. Third, while we included a number of important control variables in our analysis we did not control for all the variables that could possibly affect foreign subsidiary performance. Future studies of international market selection may wish to include other additional control variables in their normative tests of international market selection. Fourth, it is possible that a firm may only be willing to select a certain market if it can participate there through a certain mode. Based on our data set (which also includes mode choice) we did not find a correlation between the international market (host country) selected and the mode type chosen. However, future efforts may wish to examine this issue in a more comprehensive fashion. It must be noted that managers' perceptions of *Market Potential*, *Government Risk*, *Cost of Contracts*, etc. were based on an assessment of how they were at the time they filled out the questionnaire rather than what it was like when they made the IMS. Future studies may improve on this study by asking about the variables for both (1) the time that they made the selection and (2) the time of the post-entry performance analysis.

Finally, this paper was restricted to examining only market-seeking international market selection. Many reasons exist why firms choose a particular location. This can include an institutional reason such as legitimacy, strategic reasons like following your competitor (Knickerbocker, 1973), being present in a competitor's key home market (Hamel & Prahalad, 1985) and/or for resource-seeking or asset seeking reasons. Future efforts may wish to explore whether the OLI model is predictive and normative with respect to differing motivations.

Future research efforts may wish to focus on a number of other issues. The effectiveness of the eclectic framework for IMS decisions in other regions of the world or the OLI variables in an industry specific context or the motives governing the selection process itself (e.g. profit maximization versus growth in market share) could be investigated. Questions to be explored might include: Does the nature and level of competition moderate the influence of the OLI variables on IMS and if so, how? Future work in this area might focus on the provision of analytical tools based on the eclectic framework that could function as practical templates for future IMS decisions. Finally, our paper is restricted to first-time entries into the host country market. Future studies examining subsequent investments in the same host country market may need to add a *firm's experience with the focal host country* variable in order to assess the impact of country-specific experience.

Researchers may also wish to explore the eclectic framework from the perspective of host country governments that are attempting to attract market-seeking investment. What, if any, is the relationship between government policies and the OLI framework? To what extent, if any, do government inducements complement OLI variables? It is also possible that a firm may be willing to select a certain market only if it can participate there through a particular mode. While such speculation is beyond the scope of our study, in order to address this issue future efforts will need to ask: is a firm willing to select a certain market in which to participate only if it can participate there through a specific mode type?

In this paper we attempted to add new insights to the international market selection literature by extending an existing model to predict firms' choices of international markets. We found that Dunning's OLI framework provided both a predictive and normative model of IMS. As Singh and Kundu (2002) suggest, the OLI framework is a static model; it ignores the dynamic aspects of the IMS decision. Future scholarship, using other theories or placing the OLI framework in a dynamic context, may help managers and scholars better understand how strategic models can be used to improve international market selection and subsequent MNE subsidiary performance.

Appendix A. Detailed measures of constructs

Ownership advantages (short reference name)	Description	Scale
World Wide Sales (SALESWW)	Firm sales volume	Continuous ratio measure
Experience	Sales intensity	Sales outside domestic market divided by total worldwide sales
	Years experience outside the home country	Continuous ratio measure
	Number of countries outside the home country	Continuous ratio measure
R&D intensity (PERRD)	Research and development spending divided by firm sales volume	Continuous ratio measure
Creativeness rating (CREATIVE)	Perceived ability of the firm to create new products and services	7-point bi-polar scale

Location advantages (short reference name)	Description	Scale
Location1	Perceived managerial assessment of market size	7-point bi-polar scale
	Perceived managerial assessment of growth potential	7-point bi-polar scale
General stability (STABIL)	Perceived managerial stability of the general political, social and economic conditions of the host country	7-point bi-polar scale
Risk of government (RISKGOV)	Managerial perceptions of the host government's policies toward conversion and repatriation of profits	7-point bi-polar scale

Internalization advantages (short reference name)	Description	Scale
Cost of contracts (COSTCON)	Compared to their home country how managers perceive the costs of making and enforcing contracts	7-point bi-polar scale
Risk of dissemination (RISKDIS)	Perceived managerial risk of dissipation of proprietary knowledge	7-point bi-polar scale

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